



CONSORT RANDOMIZED CLINICAL TRIAL

Effect of Foraminal Enlargement on Postoperative Pain in Necrotic Single-Rooted Teeth: A Randomized Clinical Trial

Francisca Soares Meire de Freitas
Portela, DDS, MSc, PhD,*
Alexandre Sigrist De Martin, DDS,
MSc, PhD,*
Rina Andréa Pelegrine, DDS,
MSc, PhD,* James L. Gutmann,
DDS, MSc, PhD,[†]
Augusto Shoji Kato, DDS, MSc,
PhD,[‡] and
Carlos Eduardo da Silveira Bueno,
DDS, MSc, PhD*

SIGNIFICANCE

Foraminal enlargement was related to higher levels of pain than foraminal patency after performing root canal therapy with WaveOne Gold instruments and also to higher rates of sealer extrusion. Nevertheless, no significant correlation between extrusion and postoperative pain was observed.

From the *Faculdade São Leopoldo Mandic, Instituto de Pesquisa São Leopoldo Mandic, Endodontia, Campinas, São Paulo, Brazil; [†]Department of Endodontics, Nova Southeastern University, College of Dental Medicine, Davie, Florida; and [‡]Department of Restorative Dentistry, Endodontics, and Dental Materials, Discipline of Endodontics, School of Dentistry of Bauru, University of São Paulo, Bauru, São Paulo, Brazil

Address requests for reprints to Dr Carlos Eduardo da Silveira Bueno, Faculdade São Leopoldo Mandic, Instituto de Pesquisa São Leopoldo Mandic, Endodontia, Rua Dr José Rocha Junqueira, 13, Campinas, SP, Brazil 13045-755.
E-mail address: carlosesbueno@terra.com.br
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KEY WORDS

Foraminal enlargement; postoperative pain; reciprocating movement; root canal therapy; WaveOne Gold system

Several factors have been related to postoperative pain after root canal therapy of asymptomatic teeth, including patient age and sex, tooth type, previous pain, and apical radiolucency^{1,2}. Previous studies have also assessed the relationship of postoperative pain and the use of different instrumentation systems^{3,4}. Reciprocating systems have been shown to promote faster canal shaping with fewer instruments while maintaining the original path of the canal^{5,6}. Furthermore, although the extrusion of pathogens has been linked to postoperative pain in nonvital teeth, the use of reciprocating instruments has been proven not to increase the apical extrusion of debris^{7,8}.

The WaveOne Gold system (Dentsply Maillefer, Ballaigues, Switzerland) has a parallelogram-shaped cross section with 2 cutting edges and an off-center design feature. The nickel-titanium alloy used in its production ("Gold") undergoes a thermal treatment through which each instrument is heated and then cooled slowly, a procedure that reportedly improves instrument elasticity⁹; in contrast, production of the M-Wire nickel-titanium alloy of its predecessor involves premanufacturing thermal processing. However, to date, it remains unknown whether these differences could influence treatment outcomes with respect to postoperative pain.

Bacteria found on the external surface of tooth roots are claimed to maintain radiolucency and apical periodontitis¹⁰. Because biofilm in the apical portion of the root canal can be difficult to access and

remove, some authors recommend performing foraminal enlargement (FE) ^{11,12}. This additional procedural step has been shown to enhance periapical healing in animal models ^{13–15}. Furthermore, FE differs from just maintaining foraminal patency (FP) in that its goal is to clean and shape the entire extent of the canal, whereas the goal of the latter procedure is to pass a small file passively through the apical foramen, merely to avoid blockage.

Although FE has been shown to improve disinfection of the apical area, where ramifications capable of harboring bacteria are abundant ¹⁶, it has also been found to promote increased rates of deviation of the foramen ¹⁷. On the other hand, a recent literature review found moderate evidence that treatments performed with or without maintaining FP were associated with similar levels of pain ¹⁸.

The aim of the present study was to assess the perception of pain, the use of pain relief medication, and the extrusion of filling material related to single-visit root canal treatment of asymptomatic necrotic single-rooted teeth with or without FE. The correlation between sealer extrusion and pain was also investigated. The null hypothesis tested was that there would be no difference between teeth treated with or without FE with respect to the variables assessed.

MATERIALS AND METHODS

This prospective randomized clinical study was conducted in the clinical setting of the private practice of 1 of the researchers (F.S.M.F.P.) and was approved by the local research ethics committee (approval #2.913.225). Patients taking part in the study were 19–55 years of age, had signed a free and informed consent form, and were treated from February 2019 to January 2020. Sample size calculation was based on a previous study ³ and was performed using the analysis of variance test. Considering a difference among the mean results of 0.03, a standard deviation of 0.05, a power of 0.80, and an alpha error of 0.05, the calculation yielded a sample size requirement of 69 patients per group. Seventy patients were enrolled in each experimental group to allow for 1 loss to follow-up ($n = 70$).

The inclusion criteria were asymptomatic, necrotic teeth with a single, fully formed and straight root canal with or without apical radiolucency and a maximum periodontal probing depth of 3 mm. Only patients whose general health status was classified as American Society of Anesthesiologists class I were included in the study. Patients with any chronic or acute systemic disease; who had taken antibiotics or

painkillers in the last 10 days; and had teeth associated with any sign of swelling, sinus tract, root resorption (internal or external), or previous endodontic treatment (assessed radiographically) were excluded. At the time of treatment, patients were asked to randomly select a sealed envelope containing a paper slip indicating allocation to either the FP or FE group (single-blind design).

Patients were anesthetized with 3.6 mL 2% lidocaine with 1:100,000 epinephrine solution, either by buccal infiltration or inferior alveolar nerve block for maxillary or mandibular teeth, respectively. After rubber dam isolation and surgical field decontamination, the pulp chamber of the tooth was reached with a round diamond bur. Any sign of bleeding led to the removal of the tooth from the study (none occurred).

Copious irrigation was performed with 5 mL 2.5% sodium hypochlorite (NaOCl) solution using a syringe (Becton & Dickson, Curitiba, PR, Brazil) and a 24-G needle (Ultradent, South Jordan, UT), after which the root canal was carefully inspected for initial content removal and confirmation of a single canal. A WaveOne Gold file (Dentsply Maillefer) was selected, and the cervical and middle thirds of the root canal were instrumented following the manufacturer's instructions. The apical foramen was located with an electronic apex locator (Propex II, Dentsply Sirona, York, PA), and the working length (WL) was set either at the foramen level ("0.0" reading on the electronic apex locator) or at 1 mm short of the foramen in the FE and FP groups, respectively. Root canal shaping was performed with an appropriately sized instrument coupled to an X-Smart Plus engine (Dentsply Maillefer) using in-and-out movements up to the WL established for each group, as described previously. After each set of 3 movements of 3–4 mm in both groups, the instrument was removed and cleaned, a #10 K-file was used to check for canal patency, and the canal was irrigated with 2.5 mL 2.5% NaOCl using a 5-mL disposable syringe and a 24-G hypodermic needle positioned 3 mm short of the WL. The canals were instrumented either up to size 25/.07 ("Primary"), when the initial apical diameter corresponded to a #15 or #20 file, or up to size 45/.05 ("Large"), when the initial apical diameter corresponded to a #25 file or higher.

After instrumentation, the final irrigation was performed using an ultrasonic insert (Irrisonic; Helse Technology, Santa Rosa do Viterbo, SP, Brazil) positioned 2 mm short of the WL and driven by an ultrasonic device (EMS, São Paulo, SP, Brazil) set to operate at a power of 10%. The canal was initially irrigated with 3 mL 17% EDTA solution in three 1-mL agitation cycles of 20 seconds, then with 6 mL

2.5% NaOCl in three 2-mL cycles of 20 seconds, and lastly with a final 5-mL rinse of 2.5% NaOCl, totaling 25 mL 2.5% NaOCl per specimen. The canal was then aspirated using a capillary tip (Ultradent) and dried with absorbent paper cones.

Afterward, the canal was filled with a gutta-percha cone corresponding to the last instrument fitted to the WL using the single-cone technique. The cone was coated with AH Plus sealer (Dentsply Maillefer), and then placed inside the canal to provide a 3-dimensional sealing effect. The coronal portion of the filling material was removed with a heated plugger followed by cold vertical condensation. Finally, a 2-mm layer of Coltosol (Coltene/Vigodent, Rio de Janeiro, RJ, Brazil) was applied to seal the canal orifice. Subsequently, provisional restoration of the pulp chamber was performed using glass ionomer cement. Finally, the occlusion was adjusted, and a periapical radiograph was taken to check for sealer extrusion (Fig. 1A and B'). All of the procedures were performed by 2 endodontists with more than 10 years of experience who were trained to perform a standardized treatment protocol. Furthermore, at the end of the training session, both endodontists were given a printed checklist summarizing all of the steps of the protocol. A dental operating microscope (Alliance Microscopia, São Carlos, SP, Brazil) was used at 10× magnification during all of the clinical procedures by both endodontists.

The patients were instructed to make an appointment with the Department of Restorative Dentistry for final restoration procedures and to take 400 mg ibuprofen every 6 hours in the event that they needed pain relief (none of the patients were allergic to this medication as per the anamnesis). Patients were also given 4 copies of a pain evaluation instrument and were instructed to record their pain sensation 24, 48, and 72 hours and 1 week postoperatively. They were also informed that they could contact the clinician in the event that they needed additional care to address any tooth-related emergency.

The evaluation instrument was a modified visual analog scale, with scores ranging from 0–10, which were grouped into the following categories: no pain (0), mild pain (1–3), moderate pain (4–7), and severe pain (8–10). The visual analog scale also included schematic drawings of facial expressions reflecting the different pain perceptions to aid patients in selecting the appropriate score.

A dental assistant not involved in the treatment procedures contacted each patient by phone at each assessment time point of the study and entered the information collected in a spreadsheet. In the same interview, patients

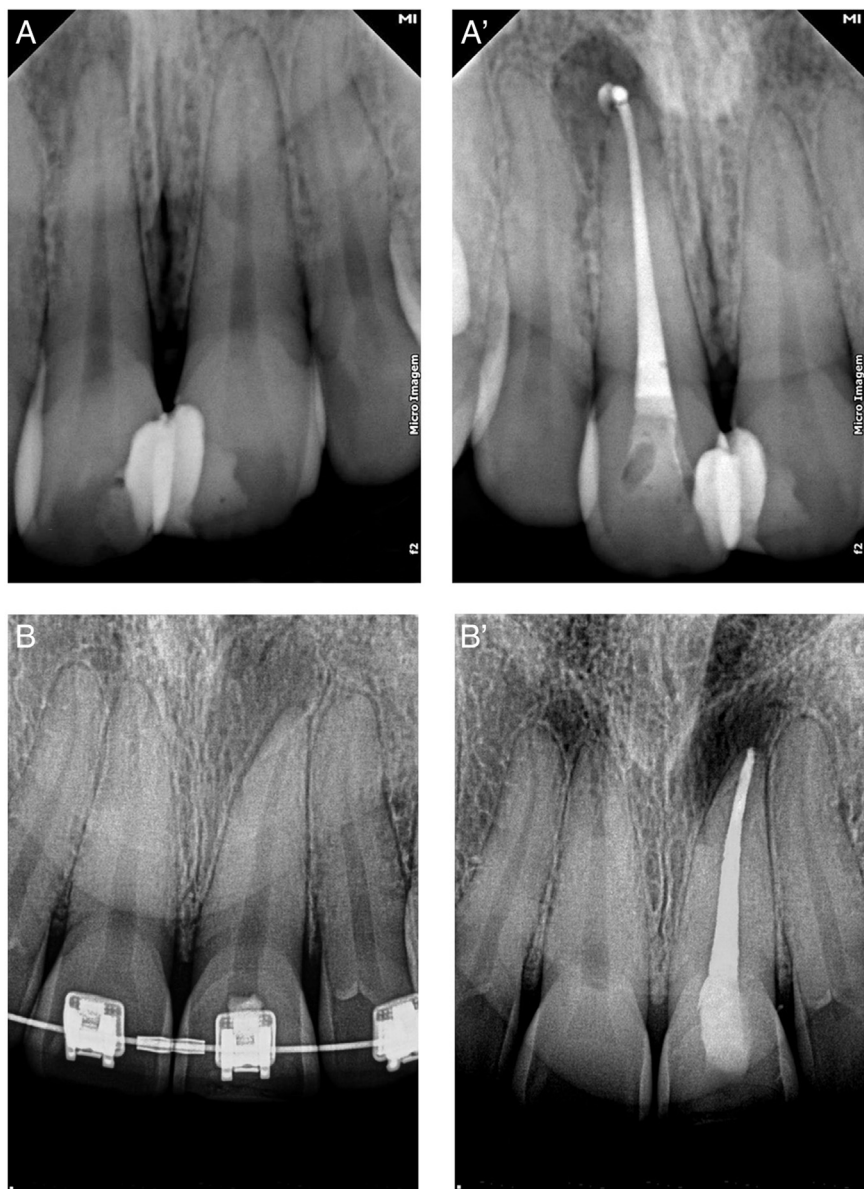


FIGURE 1 – Pre- and postoperative radiographs representative of the cases treated in the study. (*A* and *A'*) The FE group and (*B* and *B'*) the FP group.

who had recorded any pain score were asked whether or not they had taken any medication to relieve pain, and, if so, how many tablets.

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